



DeepLearning.AI

RAG Evaluation

Retrieval Quality Metrics

Common ingredients to most retriever quality metrics:

The Prompt

The specific
prompt being
evaluated

Ranked Results

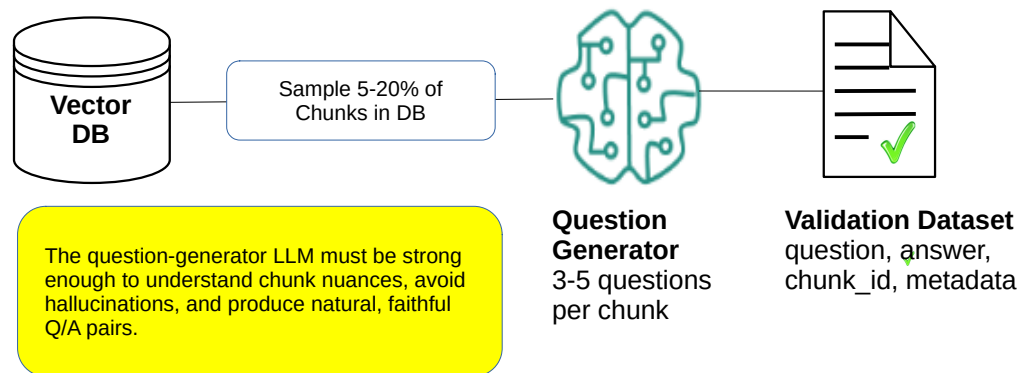
Documents
returned in
ranked order

Ground Truth

All documents
labeled as
relevant or
irrelevant

If you want to evaluate your retriever
you need to know the correct answers

Generating a Validation/Test set



Example:

Vector DB has 1000 chunks

Select 10%→100 chunks

Generate 5 questions per chunk

Have a validation set of $5 \times 100 = 500$ question/chunk pairs

*A consequence of this is that you only have 1 relevant chunk per question

Precision and Recall

Precision

Measures how many of the returned documents are relevant

$$\text{Relevant Retrieved} / \text{Total Retrieved}$$

Recall

Measures how many of the relevant documents are returned

$$\text{Relevant Retrieved} / \text{Total Relevant}$$

Example



10 Relevant Documents in Knowledge Base



First Run

Retrieved: 12 Documents

Relevant: 8 Documents

Precision (8/12)

66%

Recall

(8/10)

80%



Second Run

Retrieved: 15 Documents

Relevant: 9 Documents

Precision (9/15)

60%



Recall

(9/10)

90%

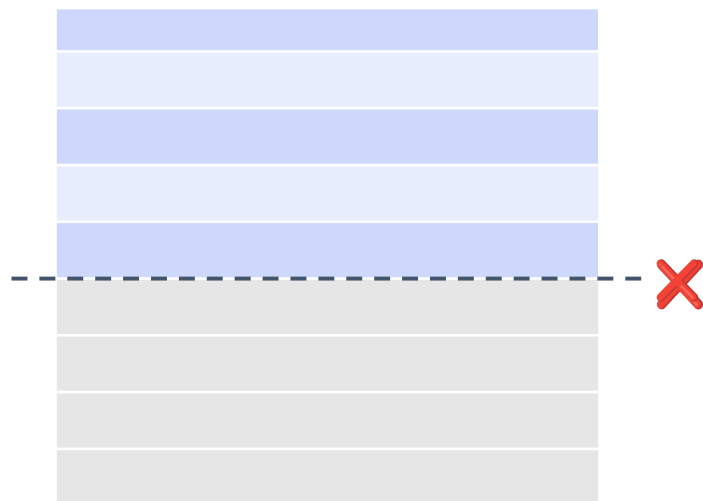


Precision penalizes for **returning irrelevant** documents

Recall penalizes for **leaving out relevant** documents

100% Precision & Recall: Rank relevant documents most highly and only return those

Top k



Top K

- Retrieval metrics are influenced by **how many documents the retriever returns**
- Metrics are discussed in terms of **top-k** documents

Example

For a given query a vector DB has 8 relevant chunks

Precision = relevant retrieved / total retrieved

Recall = relevant retrieved / total relevant

Rank	Relevant
1	relevant
2	-
3	-
4	relevant
5	-
6	relevant
7	relevant
8	-
9	relevant
10	relevant

Precision @5

2 out of 5

40%

Precision @10

6 out of 10

60%

Recall@10

6 out of 8 total
relevant

75%

Top-5 or Top-1 is stricter

Top-5 to Top-15 often used

What if there is only 1 relevant chunk per query?

If it's returned in top K?

$$P@5 = 1/5 = .2$$

$$P@10 = 1/10 = .1$$

Rank	Relevant
1	relevant
2	-
3	-
4	-
5	-
6	-
7	-
8	-
9	-
10	-

Top 5 {

{ Top 10

What if there is only 1 relevant chunk per query?

If it's **not** returned in top K?

$$P@5 = 0/5 = 0$$

$$P@10 = 0/10 = 0$$

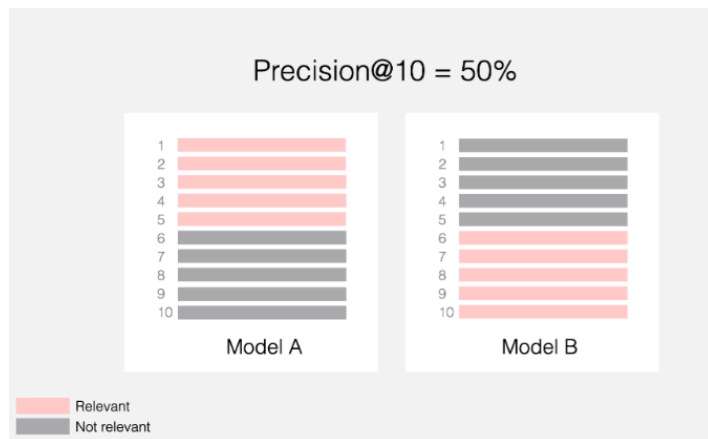
	Rank	Relevant	
Top 5	1	-	Top 10
	2	-	
	3	-	
	4	-	
	5	-	
	6	-	
	7	-	
	8	-	
	9	-	
	10	-	

Precision @k, Pros and Cons

Easy to explain: Precision@K is intuitive, so it's good for communicating system quality to stakeholders.

Tracks accuracy (not order): It measures how many of the top-K results are relevant, regardless of their positions.

Not rank-aware: It ignores where relevant items appear—5 relevant in positions 1–5 and 6–10 both give 50% at K=10.



Same performance
Even though left selection
ordered relevant chunks
higher in list

Summary: Use Precision@K when you want the accuracy of the top-K results without caring about their order.

Mean Average Precision

MAP@K evaluates average precision for relevant documents in first K documents. It is built off a related metric called “average precision”.

Rank	Item	Precision@K
1	relevant	1/1 1.0
2	-	1/2 0.5
3	-	1/3 0.3
4	relevant	2/4 0.5
5	relevant	3/5 0.6
6	-	3/6 0.5

Rewards ranking relevant documents highly

1

Sum precisions for relevant docs only.

$$1 + 0.5 + 0.6 = 2.1$$

2

Divide by number of relevant documents

$$2.1 / 3 = 0.7$$

3

This calculation gives **Average Precision** or **AP**, for Mean Average Precision you find the average AP value across many prompts

Reciprocal rank

Measures the rank of the first relevant document in the returned list

$$\text{Reciprocal Rank} = 1 / \text{Rank}$$

First relevant at rank 1

1.0

First relevant at rank 2

0.5

First relevant at rank 3

0.33

No relevant docs = $1/\infty$

0.0

The later the first relevant document appears, the worse the reciprocal rank

Mean Reciprocal Rank (MRR) averages over many prompts

Mean Reciprocal Rank

Search
1

First relevant at rank 1

1.0

Search
2

First relevant at rank 3

0.33

Search
3

First relevant at rank 6

0.17

Search
4

First relevant at rank 2

0.5

1 **Sum all ranks**

$$1.0 + 0.33 + 0.17 + 0.5 = 2.0$$

2 **Divide by number of searches**

$$2.0 / 4$$

$$\mathbf{MRR = 0.5}$$

How to use retriever metrics

Recall or recall@K

Most cited metric, captures fundamental goal of finding relevant documents

Precision & MAP

Asses irrelevant documents and ranking effectiveness

Mean Reciprocal Rank

How well model performs at the very top of ranking

Metrics help:

- Evaluate retriever performance
- Check if adjustments improve results

**All metrics depend on having
ground truth relevant
documents**